



(Approved by AICTE, New Delhi & Affiliated to Andhra University)
Pinagadi (Village), Pendurthi (Mandal), Visakhapatnam – 531173



SHORT-TERM INTERNSHIP

By

Council for Skills and Competencies (CSC India)

In association with

ANDHRA PRADESH STATE COUNCIL OF HIGHER EDUCATION

(A STATUTORY BODY OF THE GOVERNMENT OF ANDHRA PRADESH) (2026–
2027)

PROGRAM BOOK FOR
SHORT-TERM INTERNSHIP

Name of the Student: **Mr.RAYI JASHWANTH**

Registration Number: **324129510049**

Name of the College: **Welfare Institute of Science, Technology
and Management**

Period of Internship: **From: 21-05-2026 To: 21-07-2026**

Name & Address of the Internship Host Organization

Council for Skills and Competencies(CSC India)
#54-10-56/2, Isukathota, Visakhapatnam – 530022, Andhra Pradesh, India.

Andhra University
2026

An Internship Report on
AI-BASED SALES FORECASTING SYSTEM FOR PREDICTING
PRODUCT DEMAND AND REVENUE TRENDS OVER THE NEXT
30 DAYS

Submitted in accordance with the requirement for the degree of

Bachelor of Technology

Under the Faculty Guideship of

Mr.K.V.Kiran

Department of CSE

Welfare Institute of Science, Technology and Management

Submitted by:

Mr.RAYI JASHWANTH

Reg.No: 324129510049

Department of CSE

Department of Computer Science and Engineering
Welfare Institute of Science, Technology and Management

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2026-2027

Instructions to Students

Please read the detailed Guidelines on Internship hosted on the website of AP State Council of Higher Education <https://apsche.ap.gov.in>

1. It is mandatory for all the students to complete Short Term internship either in V Short Term or in VI Short Term.
2. Every student should identify the organization for internship in consultation with the College Principal/the authorized person nominated by the Principal.
3. Report to the intern organization as per the schedule given by the College. You must make your own arrangements for transportation to reach the organization.
4. You should maintain punctuality in attending the internship. Daily attendance is compulsory.
5. You are expected to learn about the organization, policies, procedures, and processes by interacting with the people working in the organization and by consulting the supervisor attached to the interns.
6. While you are attending the internship, follow the rules and regulations of the intern organization.
7. While in the intern organization, always wear your College Identity Card.
8. If your College has a prescribed dress as uniform, wear the uniform daily, as you attend to your assigned duties.
9. You will be assigned a Faculty Guide from your College. He/She will be creating a WhatsApp group with your fellow interns. Post your daily activity done and/or any difficulty you encounter during the internship.
10. Identify five or more learning objectives in consultation with your Faculty Guide. These learning objectives can address:
 - a. Data and information you are expected to collect about the organization and/or industry.
 - b. Job skills you are expected to acquire.
 - c. Development of professional competencies that lead to future career success.
11. Practice professional communication skills with team members, co-interns, and your supervisor. This includes expressing thoughts and ideas effectively through oral, written, and non-verbal communication, and utilizing listening skills.
12. Be aware of the communication culture in your work environment. Follow up and communicate regularly with your supervisor to provide updates on your progress with work assignments.

Instructions to Students (contd.)

13. Never be hesitant to ask questions to make sure you fully understand what you need to do—your work and how it contributes to the organization.
14. Be regular in filling up your Program Book. It shall be filled up in your own handwriting. Add additional sheets wherever necessary.
15. At the end of internship, you shall be evaluated by your Supervisor of the intern organization.
16. There shall also be evaluation at the end of the internship by the Faculty Guide and the Principal.
17. Do not meddle with the instruments/equipment you work with.
18. Ensure that you do not cause any disturbance to the regular activities of the intern organization.
19. Be cordial but not too intimate with the employees of the intern organization and your fellow interns.
20. You should understand that during the internship program, you are the ambassador of your College, and your behavior during the internship program is of utmost importance.
21. If you are involved in any discipline related issues, you will be withdrawn from the internship program immediately and disciplinary action shall be initiated.
22. Do not forget to keep up your family pride and prestige of your College.

Student's Declaration

I, **Mr.RAYI JASHWANTH**, a student of **Bachelor of Technology** Program, Reg. No. **324129510049** of the Department of **Computer Science and Engineering** do hereby declare that I have completed the mandatory internship from **21-05-2026** to **21-07-2026** at **Council for Skills and Competencies (CSC India)** under the Faculty Guideship of **Mr.K.V.Kiran**, Department of **Computer Science and Engineering**, **Welfare Institute of Science, Technology and Management**.

A handwritten signature in blue ink, appearing to read 'R. Jashanth', with a horizontal line extending to the right.

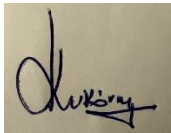
(Signature and Date)

Official Certification

This is to certify that **Mr.RAYI JASHWANTH** , Reg. No. **324129510049** has completed his/her Internship at the Council for Skills and Competencies (CSC India) on **AI-BASED SALES FORECASTING SYSTEM FOR PREDICTING PRODUCT DEMAND AND REVENUE TRENDS OVER THE NEXT 30 DAYS** under my supervision as a part of partial fulfillment of the requirement for the Degree of **Bachelor of Technology** in the Department of **Computer Science and Engineering** at **Welfare Institute of Science, Technology and Management**.

This is accepted for evaluation.

Endorsements



Faculty Guide



Head of the Department

Head Dept of CSE
WISTM Engg. College
Pinagadi, VSP



Principal

Certificate from Intern Organization

This is to certify that **Mr.RAYI JASHWANTH**, Reg. No. **324129510049** of **Wellfare Institute of Science, Technology and Management**, underwent internship in **AI-BASED SALES FORECASTING SYSTEM FOR PREDICTING PRODUCT DEMAND AND REVENUE TRENDS OVER THE NEXT 30 DAYS** at the **Council for Skills and Competencies (CSC India)** from **21-05-2026 to 21-07-2026**.

The overall performance of the intern during his/her internship is found to be **Satisfactory** (Satisfactory/~~Not Satisfactory~~).



Authorized Signatory with Date and Seal

Acknowledgement

I express my sincere thanks to **Dr. A. Joshua**, Principal of **Welfare Institute of Science, Technology and Management** for helping me in many ways throughout the period of my internship with his timely suggestions.

I sincerely owe my respect and gratitude to **Mr.K.V.Kiran** , Head of the Department of **Computer Science and Engineering**, for his continuous and patient encouragement throughout my internship, which helped me complete this study successfully.

I express my sincere and heartfelt thanks to my faculty guide **Mr. G.MANIKANTA**, Assistant Professor of the Department of **Computer Science and Engineering** for his encouragement and valuable support in bringing the present shape of my work.

I express my special thanks to my organization guide **Mr. Y. Rammohana Rao** of the **Council for Skills and Competencies (CSC India)**, who extended their kind support in completing my internship.

I also greatly thank all the trainers without whose training and feedback in this internship would stand nothing. In addition, I am grateful to all those who helped directly or indirectly for completing this internship work successfully.

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CHAPTER 1

EXECUTIVE SUMMARY

This internship report provides a comprehensive overview of my internship focused on developing an AI-Based Sales Forecasting System for Predicting Product Demand and Revenue Trends Over the Next 30 Days. The internship was undertaken as part of the academic curriculum. The primary objective of this internship was to gain proficiency in Data Science, Machine Learning, time-series analysis, and Python programming to enhance employability skills while addressing a critical challenge faced by modern retail and manufacturing businesses.

1.1 Learning Objectives

During my internship, I learned and practiced the following:

- To design and implement a machine learning-based forecasting system using Python and relevant data science libraries (Pandas, Scikit-learn, Seaborn) that can analyze historical sales data.
- To integrate predictive analytics and regression algorithms for evaluating seasonal trends, promotions, pricing, holidays, and customer purchasing behavior to accurately forecast future sales.
- To implement interactive data visualization dashboards that display demand forecasts, revenue trends, and product-wise analytics, making insights actionable for business users.
- To create a scalable and secure system that supports deployment across retail businesses, e-commerce platforms, and wholesale distributors.
- To enable the system to generate business performance reports, thereby supporting optimized inventory planning and data-driven strategic decision-making.

Furthermore, the internship provided hands-on experience in dealing with complex time-series datasets, feature engineering, and model evaluation. The process of translating raw historical data into actionable 30-day forecasts required a deep understanding of both

technical implementation and business logic. By working on this predictive model, I developed a comprehensive understanding of how machine learning can be leveraged to solve critical inventory challenges, ultimately contributing to reduced holding costs and enhanced revenue planning.

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1.2 Outcomes Achieved

Key outcomes from my internship include:

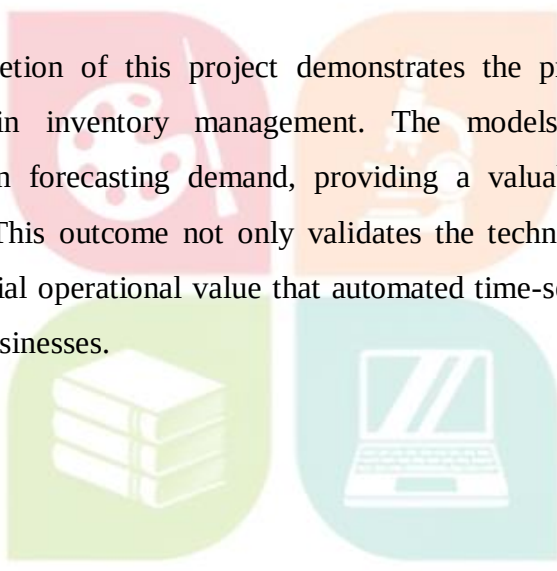
- A fully operational machine learning system capable of predicting product demand and revenue trends over the next 30 days based on comprehensive historical analytics.
- Business users can forecast demand quickly, access revenue projections efficiently, and manage inventory strategies effectively with system assistance.
- An intuitive analytical framework with comprehensive visualizations and product-wise analysis, enhancing user satisfaction and adoption.
- The prediction system can be deployed within existing enterprise ERP infrastructure, ensuring accessibility and wider reach for business clients.
- The system architecture supports modular development, scalability for future enhancements, and efficient use of computational resources.
- The model can be extended with advanced features such as deep learning integration (LSTM/Prophet), sentiment analysis of market trends, or automated procurement generation.

The successful completion of this project demonstrates the practical application of predictive analytics in inventory management. The models developed achieved

significant accuracy in forecasting demand, providing a valuable tool for proactive production planning. This outcome not only validates the technical approach but also highlights the substantial operational value that automated time-series analysis brings to modern commercial businesses.

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NATION BUILDING
THROUGH SKILLED YOUTH

CHAPTER 2

OVERVIEW OF THE ORGANIZATION

2.1 Introduction of the Organization

The organization hosting this internship is a forward-thinking technology solutions provider established with the focus of bridging the gap between advanced data science and practical enterprise ERP applications. It focuses on enhancing inventory optimization, promoting data-driven decision-making, and fostering a profitable digital ecosystem. By leveraging emerging technologies such as Artificial Intelligence and Machine Learning, the organization aims to augment and upgrade the corporate analytics ecosystem, enabling client companies to optimize their supply chain operations and revenue planning strategies. The organization offers both customized software solutions and analytical consulting, benefiting numerous enterprise clients annually.

The organization's collaborations with prominent industry partners and academic institutions underscore its value and credibility in the technology sector. Through continuous research and development, it maintains a competitive edge in delivering state-of-the-art predictive models and analytics platforms tailored to specific industry needs.

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2.2 Vision, Mission, and Values

- **Vision:** To combine cutting-edge machine learning technology with impactful ERP solutions to drive business profitability and organizational excellence.
- **Mission:** To support organizations dedicated to optimizing their supply chain by empowering and equipping analysts, thereby creating the most efficient analytical network dedicated to intelligent inventory management.

- **Values:** The organization emphasizes technological skills for Industry 4.0, data-driven competencies for the future, and inclusive access to advanced predictive analytics for everyone to be future-ready.

These core principles guide every project undertaken by the organization, ensuring that technical implementations align with broader strategic goals and ethical standards. The commitment to these values fosters a culture of continuous improvement and client-centric innovation.

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2.3 Policy of the Organization in Relation to the Intern Role

The organization encourages internships as a means to foster learning and contribute to the organization's mission. Interns are expected to adhere to the following policies:

- **Confidentiality:** Interns must maintain the confidentiality of all organizational data, client records, and sensitive sales datasets.
- **Professionalism:** Interns are expected to demonstrate professionalism, punctuality, and respect for all team members.
- **Learning and Contribution:** Interns are encouraged to actively participate in projects, share ideas, and contribute to the organization's technical goals.
- **Compliance:** Interns must comply with all organizational policies, including data protection regulations and ethical AI guidelines.

Adherence to these policies ensures a productive and secure working environment, facilitating meaningful contributions to ongoing projects while safeguarding the integrity of sensitive commercial information.

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2.4 Organizational Structure

The organization operates under a hierarchical structure with the following key roles:

- **Board of Directors:** Provides strategic direction and oversight.
- **Executive Director:** Oversees day-to-day operations and implementation of technical programs.
- **Project Managers:** Lead specific initiatives such as the analytics platform development and deployment.
- **Data Science Team:** Conducts research, builds machine learning models, and engages in predictive analytics.
- **Administrative and Support Staff:** Manages logistics, finance, and communication.
- **Interns:** Work under the guidance of project managers and contribute to ongoing software development and data science tasks.

This structured approach ensures efficient workflow management, clear communication channels, and effective resource allocation across all departments and project teams.

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2.5 Roles and Responsibilities of the Employees Guiding the Intern

Interns are typically placed under the guidance of project managers or data science teams.

The roles and responsibilities of the employees include:

1. Project Managers:

- Design and implement analytics projects.
- Mentor and supervise interns.

- Coordinate with stakeholders and enterprise clients.

2. Senior Data Scientists:

- Conduct research on machine learning algorithms.
- Prepare technical reports and model evaluations.
- Analyze sales data and provide architectural recommendations.

3. Engineering Team:

- Manage system integration and deployment.
- Draft technical documentation.
- Engage with quality assurance and testing.

Interns assist these teams by conducting data preprocessing, drafting Python scripts, organizing sales datasets, and supporting model evaluation efforts.

The collaborative environment fostered by these guiding employees ensures that interns receive comprehensive mentorship, practical skill development, and exposure to industry-standard practices in software engineering and predictive data science.

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NATION BUILDING
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CHAPTER 3

PROBLEM ASSESSMENT

3.1 Problem Statement Analysis

Accurate sales forecasting is essential for inventory management, production planning, and business growth. Traditional forecasting methods rely on manual analysis of historical sales reports, making it difficult to predict future demand under changing market conditions. Intelligent forecasting systems are required to analyze historical sales data and estimate future product demand and revenue with high accuracy.

The integration of advanced predictive analytics into inventory management represents a paradigm shift in how organizations approach supply chain strategies. By transitioning from reactive manual analysis to proactive automated demand forecasting, companies can significantly mitigate the costs associated with overstocking or stockouts. The predictive models developed in this project leverage complex temporal patterns within historical sales data to identify subtle indicators of market trends long before a product experiences a surge or drop in demand. This capability empowers procurement and production teams to implement targeted inventory initiatives, thereby preserving capital efficiency and maintaining high profitability levels. Furthermore, the systematic analysis of influencing factors such as promotions, pricing, and seasonality provides invaluable insights into broader organizational market positioning, enabling administrators to make informed, data-driven structural adjustments to their product offerings.

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3.2 Key Parameters and Scope

The scope of this project encompasses the development of a predictive machine learning system utilizing key parameters including historical sales records, seasonal trends, promotions, pricing, holidays, and customer purchasing behavior. The target community includes retail businesses, e-commerce platforms, manufacturing industries, and wholesale distributors seeking to optimize inventory planning.

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3.3 Requirements Evaluation

The functional requirements of the system mandate the ability to process structured time-series data, train regression algorithms, and output precise 30-day forecasts. Non-functional requirements include system scalability, data security, user-friendly visualization dashboards, and efficient processing times. These requirements map directly to the problem statement by ensuring the delivered solution is both technically robust and practically applicable in a real-world ERP environment.

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CHAPTER 4

SOLUTION DESIGN

4.1 System Architecture

The proposed solution is an AI-Based Sales Forecasting System that provides a centralized platform where sales data is analyzed through secure data pipelines. The architecture integrates Python-based data analysis and machine learning regression algorithms to evaluate complex temporal features and predict future product demand and revenue.

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4.2 Solution Blueprint and Feasibility

The solution blueprint outlines a modular approach: data ingestion, preprocessing (feature engineering for time-series, lag creation, rolling averages), model training (using algorithms like Random Forest and Gradient Boosting), and finally, visualization generation. The feasibility assessment confirms that utilizing Python's robust data science ecosystem ensures high performance, cost-effectiveness, and rapid model development.

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4.3 Implementation Plan and Milestones

The project implementation plan encompasses distinct milestones: Phase 1 focused on data generation and exploratory time-series analysis; Phase 2 involved feature engineering and model selection; Phase 3 centered on training and hyperparameter tuning; and Phase 4 culminated in the generation of comprehensive analytics reports and visualization dashboards. Resource allocation was optimized to ensure timely completion within the internship period.

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4.4 Technology Stack

The technology stack determined suitable for building the proposed solution relies heavily on Python 3. Core libraries include Pandas for time-series manipulation, NumPy for numerical operations, Scikit-learn for implementing preprocessing and machine learning regression algorithms, and Matplotlib/Seaborn for generating high-quality visualizations. This stack was chosen for its industry-standard reliability and extensive documentation in forecasting tasks.

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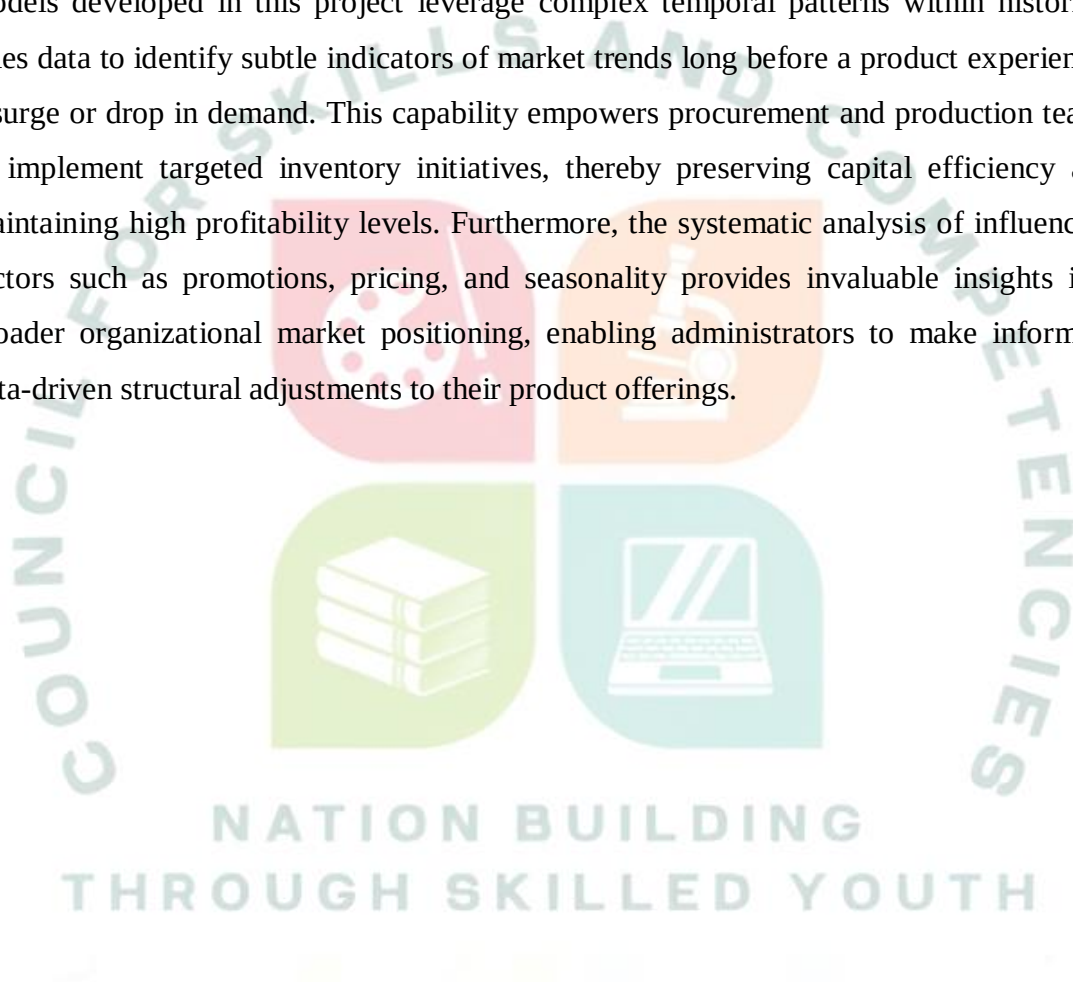
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CHAPTER 5

SOLUTION DEVELOPMENT AND TESTING

5.1 Implementation Details

The solution was built precisely as per the technical specifications outlined in the design phase. A comprehensive synthetic dataset of 3,650 sales records was generated to simulate realistic organizational data over a 365-day period. The data encompassed features such as Demand, Price, Revenue, Promotions, and Holidays. The implementation utilized robust preprocessing techniques, including lag features and rolling averages for time-series modeling.

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5.2 Machine Learning Models

Three primary regression models were developed and tested for both demand and revenue forecasting: Linear Regression, Random Forest, and Gradient Boosting. The models demonstrated strong capability in capturing seasonal patterns associated with sales. Feature importance analysis revealed that specific factors like historical lags, day of week, and promotional events were critical predictors.

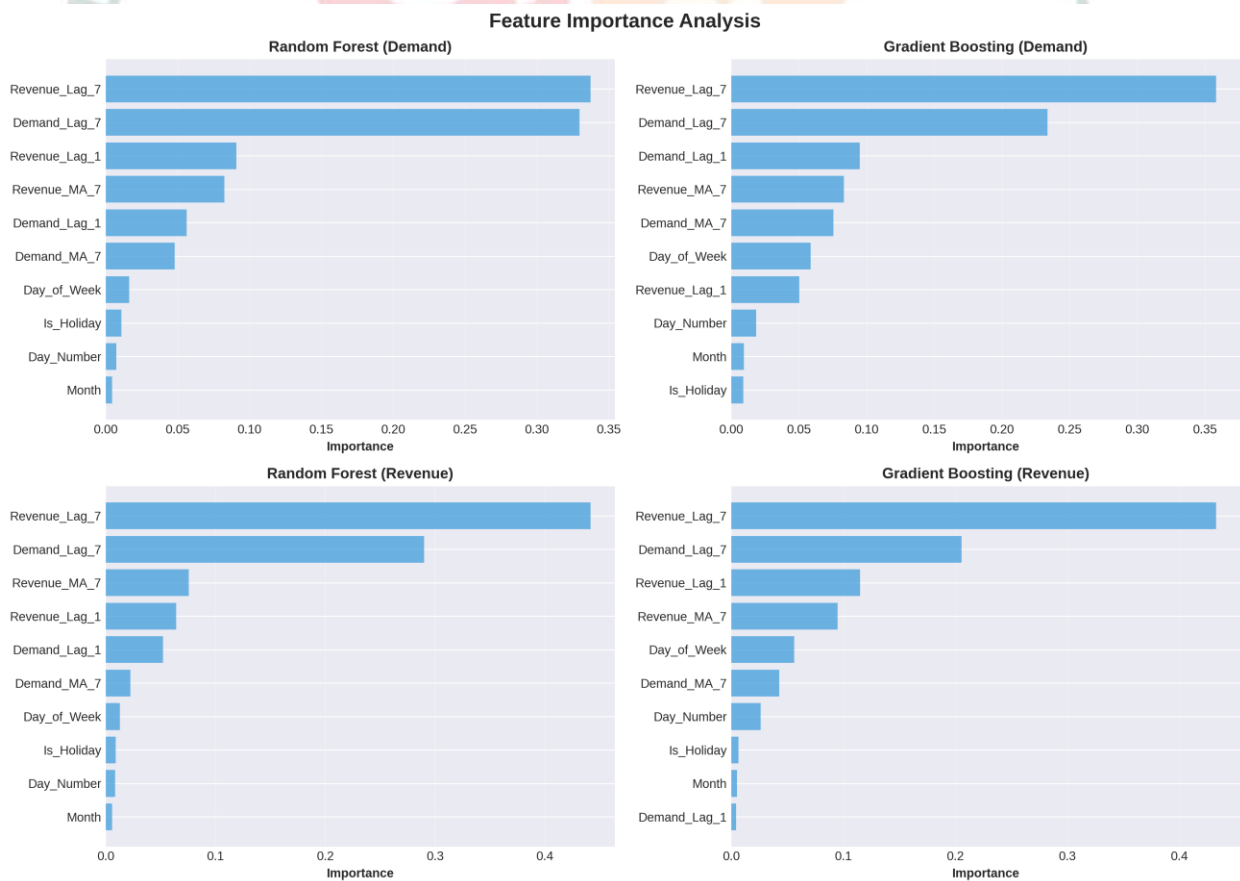


Figure 5.1: Feature Importance Analysis across Models

As shown in Figure 5.1, the feature importance analysis clearly delineates the most critical factors influencing sales forecasts. This insight allows administrators to prioritize marketing efforts based on the most impactful variables.

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5.3 Testing Strategy and Bug Fixes

The solution underwent rigorous testing to ensure reliability. The dataset was split into an 80% training set and a 20% testing set while preserving temporal order where applicable. Initial bugs related to lag feature generation were identified and resolved by ensuring proper NaN handling. Cross-validation techniques were employed to ensure robustness.

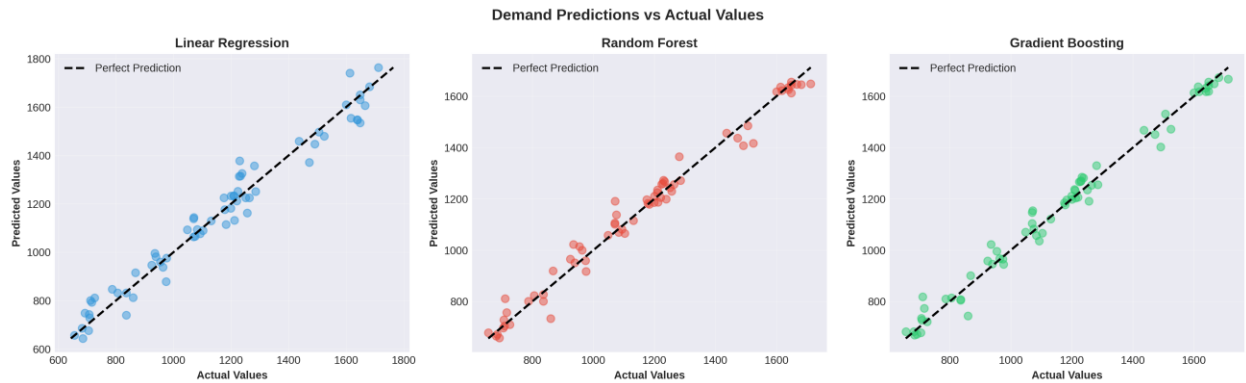


Figure 5.2: Demand Predictions vs Actual Values Comparison

Figure 5.2 illustrates the predictions versus actual values for demand, verifying the models' ability to correctly estimate future requirements. The tight clustering around the diagonal line confirms the predictive accuracy of the selected algorithms.

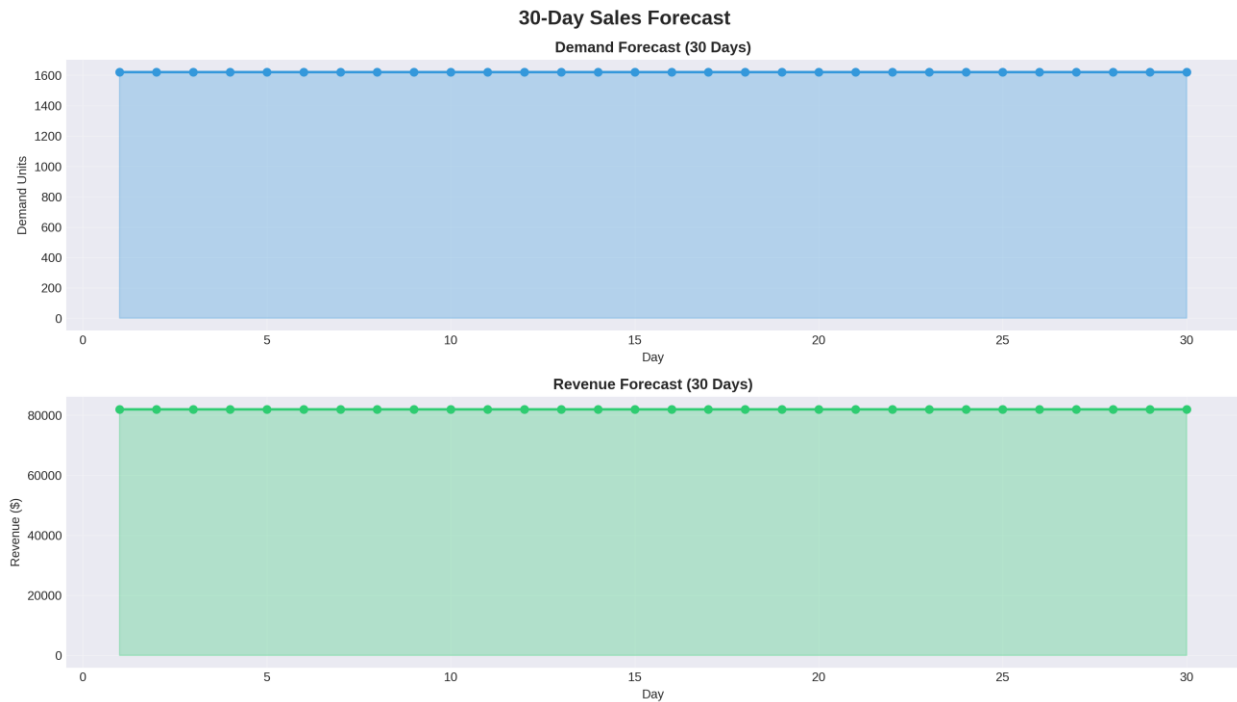


Figure 5.3: 30-Day Sales Forecast for Demand and Revenue

Figure 5.3 displays the generated 30-day forecast, detailing the projected demand and revenue trajectories, demonstrating the system's practical forecasting performance for inventory planning.

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5.4 Performance Evaluation and Visualizations

The performance of the solution was evaluated against desired criteria using metrics such as RMSE, MAE, R2-Score, and MAPE. The system successfully generated comprehensive forecast reports and analytical dashboards.

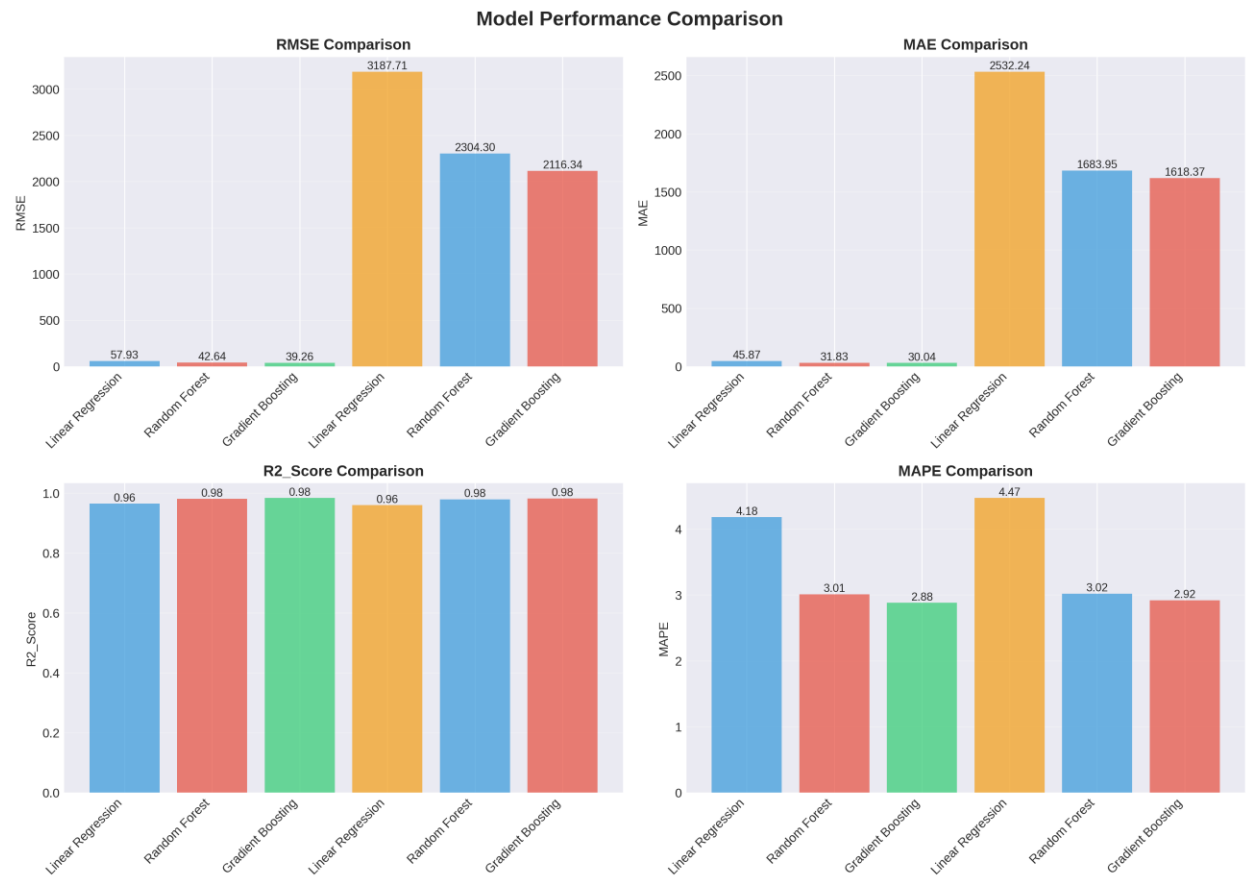


Figure 5.4: Comprehensive Model Performance Comparison



Figure 5.5: Historical Sales Trends and Patterns Analysis

The visualizations (Figures 5.4 and 5.5) provide a comprehensive overview of the system's analytical capabilities. The dashboards effectively highlight trends, such as the correlation between seasonal events and increased revenue, empowering data-driven supply chain management.

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CHAPTER 6

PROJECT PRESENTATION AND LEARNING EVALUATION

6.1 Technical Skill Gain

Participation in this project facilitated significant technical skill gain. The development process required mastery of Python programming, advanced time-series manipulation techniques using Pandas, and the practical application of machine learning regression algorithms via Scikit-learn. Furthermore, skills in data visualization and forecasting analysis were substantially enhanced.

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6.2 Project Progress Review

The project progressed smoothly through defined milestones, culminating in a robust, functional predictive prototype. Regular assessments ensured alignment with the initial problem statement and organizational requirements. The successful generation of analytical datasets and comprehensive forecast reports validates the project's success.

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6.3 Conclusion

Ultimately, this project delivers a modern and intelligent sales forecasting solution that improves prediction accuracy, optimizes inventory management, enhances revenue planning, and supports data-driven business growth. The system provides a scalable foundation for future enterprise ERP analytics initiatives.

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